

Hybrid Linear and Non-Linear Feature Selection for QSAR Classification using a Stacked Ensemble

Abstract—Predicting chemical biodegradability is a persistent challenge in quantitative structure–activity relationship (QSAR) modelling, owing to high-dimensional descriptor spaces, redundant features, and imbalanced class distributions. This study investigates the UCI QSAR Biodegradation dataset, comprising 1,055 compounds characterised by 41 molecular descriptors.

A heterogeneous stacking ensemble is proposed that pairs feature-selection methods with complementary base learners. Non-linear feature relevance is identified via Boruta and used to train a histogram-based gradient boosting classifier for capturing descriptor interactions. Concurrently, L1-regularised logistic regression selects sparse linear features for a support vector machine, while a distance-weighted k-nearest neighbours classifier operates on the full standardised feature space to exploit local structure. A regularised logistic regression meta-learner integrates predictions from all branches.

To prevent information leakage, all feature selection and model fitting are performed within a nested cross-validation framework. On an independent test set, the ensemble achieves a ROC–AUC of 0.941 and F1-score of 0.844, outperforming individual base learners on balanced metrics. These results demonstrate the value of combining diverse inductive biases through principled ensemble integration for QSAR classification.

I. INTRODUCTION

Environmental biodegradability is a key endpoint in chemical risk assessment and in regulations such as REACH [1]. Quantitative structure–activity relationship (QSAR) models provide a cost-effective alternative to experimental testing but must cope with many correlated descriptors, limited sample sizes, and class imbalance [2], [3]. Small structural changes can produce large shifts in biological activity, limiting the effectiveness of simple linear models [4]. This study investigates the UCI QSAR Biodegradation dataset [5], comprising 1,055 compounds characterised by 41 molecular descriptors.

Despite these technical challenges, accurate QSAR modelling offers significant benefits: reliable in silico predictions reduce reliance on animal testing and costly experimental assays, consistent with the 3Rs principles (Replacement, Reduction, Refinement) [6]. Such models also support the development of environmentally sustainable chemicals. However, predictions carry inherent uncertainty, and transparent reporting of limitations remains essential for responsible regulatory application.

Molecular descriptor spaces are inherently heterogeneous, characterised by both non-linear interactions and sparse linear relationships [7]. Yet many QSAR methods rely on single learners or homogeneous ensembles with uniform feature selection, failing to exploit this structural diversity. Ensemble learning provides a promising avenue for addressing these limitations by integrating complementary inductive biases [8]. Heterogeneous stacking ensembles, in particular, improve model stability and generalisation by combining diverse base learners through a meta-learning strategy [9], [10].

Several machine learning methods have proven effective for QSAR classification. Random Forest–based feature selection, such as Boruta [11], identifies relevant descriptors by comparing feature importance against randomised shadow variables. Gradient boosting methods, including histogram-based variants [12], capture complex non-linear interactions. Support vector machines remain widely used for high-dimensional classification due to their strong generalisation properties [13], while K-nearest neighbours classifiers exploit local similarity structure [14]. LASSO-based feature selection [15] provides sparse linear representations that complement non-linear approaches.

This paper introduces a heterogeneous stacking ensemble that aligns feature-selection strategies with complementary base learners. The proposed architecture pairs a histogram-based gradient boosting classifier with Boruta-selected features, a support vector machine with LASSO-selected features, and a distance-weighted K-nearest neighbours classifier operating on the full feature space. These branches are unified through an L2-regularised logistic regression meta-learner. The framework is evaluated using nested cross-validation to prevent information leakage, with performance assessed relative to individual base learners.

II. DATA PROCESSING

Prior to model training, a minimal and reproducible pre-processing pipeline was applied to ensure data integrity and eliminate redundancy. First, a basic data audit was conducted to assess dataset quality, including checks for missing values and class distribution. No missing values were detected, and the dataset showed moderate class imbalance.

Duplicate samples were then removed to prevent information leakage and artificial performance inflation. This step eliminated exact row-level duplicates, resulting in the removal of three redundant samples.

Finally, duplicate features were identified and removed by eliminating columns with identical values across all samples. This procedure reduces descriptor redundancy and multicollinearity while preserving the original feature semantics.

After pre-processing, the final dataset comprised 1,052 samples described by 41 unique features. No normalisation or transformation was applied at this stage, as feature scaling and selection were handled within the model pipelines.

III. METHODOLOGY

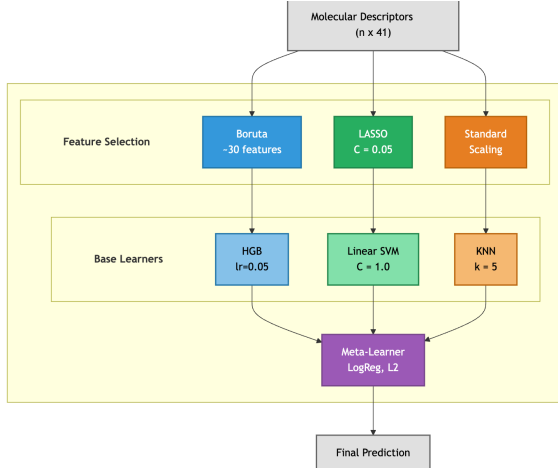


Fig. 1: Architecture of the proposed heterogeneous stacking ensemble. All steps are embedded within a nested cross-validation framework (3 outer folds, 5 inner folds).

A. Overview

This study proposes a heterogeneous stacked ensemble for QSAR classification that integrates complementary feature-selection mechanisms and learning biases. The pipeline combines (i) non-linear feature relevance detection via Boruta, (ii) sparse linear feature selection via LASSO, (iii) standard scaling for distance-based learning, and (iv) heterogeneous base learners unified through a regularised meta-learner. To avoid information leakage, all preprocessing, feature selection, and model training steps are embedded within a nested cross-validation framework

B. Dual-Pathway Feature Selection

To address the curse of dimensionality in QSAR datasets, two complementary feature-selection strategies are employed.

1) Boruta Pathway (Non-Linear Interactions):

Boruta [11] is a Random Forest-based wrapper that identifies features with predictive relevance exceeding chance. At each iteration $t \in \{1, \dots, T\}$, the design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ is augmented with shadow features $\mathbf{X}^{\text{shadow}}$, created by permuting each column of $\mathbf{X}$. A Random Forest is trained on the concatenated matrix $[\mathbf{X}, \mathbf{X}^{\text{shadow}}]$, yielding importance scores $I_j^{(t)}$ for each real feature j and $I_k^{\text{shadow},(t)}$ for each shadow feature k .

For each real feature j , a standardised score is computed:

$$Z_j = \frac{\bar{I}_j - \mu_{\text{shadow}}}{\sigma_{\text{shadow}}}, \quad (1)$$

where $\bar{I}_j = \frac{1}{T} \sum_{t=1}^T I_j^{(t)}$ is the mean importance of feature j across iterations, and μ_{shadow} and σ_{shadow} are the mean and standard deviation of all shadow feature importances.

Let $Q_\tau^{(t)}$ denote the τ -th percentile of the shadow importance distribution at iteration t . A feature j is classified as:

$$\text{status}_j = \begin{cases} \text{confirmed} & \text{if } I_j^{(t)} > Q_{90}^{(t)} \text{ for most } t \\ \text{tentative} & \text{if undecided after } T \text{ iterations} \\ \text{rejected} & \text{otherwise} \end{cases} \quad (2)$$

The selected feature set is defined as:

$$\mathcal{S}_{\text{Boruta}} = \{j : \text{status}_j \in \{\text{confirmed}, \text{tentative}\}\} \quad (3)$$

Boruta is implemented using a balanced Random Forest with $N_{\text{trees}} = 500$ and maximum depth of 5, run for $T = 100$ iterations. To ensure robustness under cross-validation, feature selection is performed once per outer training fold and subsequently frozen. The resulting subset $\mathcal{S}_{\text{Boruta}}$ is used exclusively by the non-linear base learner. In rare cases where $|\mathcal{S}_{\text{Boruta}}| < 3$ (due to single-class folds or insufficient confirmed features), a fallback selects the top-ranked features to ensure numerical stability.

Across folds, Boruta selected 32 ± 4 descriptors (mean $\pm$ std), providing a compact feature subset enriched for non-linear interactions.

2) LASSO Pathway (Sparse Linear Selection for SVM):

To obtain a sparse linear representation of the descriptor space, an L1-regularised logistic regression (LASSO) [15] is used as an embedded feature selector. Given training samples $(\mathbf{x}_i, y_i)$, the optimisation problem is:

$$\min_{\boldsymbol{\omega}} \left[\frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\text{CE}}(y_i, \hat{y}_i) + \lambda \|\boldsymbol{\omega}\|_1 \right], \quad (4)$$

where $\hat{y}_i = \sigma(\mathbf{x}_i^\top \boldsymbol{\omega})$, $\sigma(\cdot)$ denotes the logistic sigmoid, and $\mathcal{L}_{\text{CE}}$ is the binary cross-entropy loss. The regularisation parameter is set to $C = 0.05$ (corresponding to $\lambda = 1/C = 20$), and class imbalance is addressed via balanced class weights.

Features with absolute coefficients exceeding the mean coefficient magnitude are retained, forming a sparse subset $\mathcal{S}_{\text{LASSO}}$ used exclusively by the linear SVM base learner. If fewer than 5 features are selected, the top 5 by absolute coefficient are retained. A fallback mechanism retains all features when a training fold contains only one class.

C. Heterogeneous Base Learners

Two distinct base classifiers were trained using pathway-specific features, each exploiting a different inductive bias:

1) Histogram-Based Gradient Boosting (HGB):

A histogram-based gradient boosting classifier (HGB) is trained on the Boruta-selected feature subset $\mathcal{S}_{\text{Boruta}}$. This model learns an additive ensemble of decision trees by minimising a regularised empirical risk:

$$\mathcal{L} = \sum_{i=1}^n \ell(y_i, F_M(x_i)) + \lambda \sum_{m=1}^M \|f_m\|_2^2, \quad (5)$$

where $\ell(\cdot)$ denotes the logistic loss, $F_M(x)$ is the boosted prediction after M iterations, and the second term corresponds to L2 regularisation on the tree ensemble.

The learning rate is set to $\eta = 0.05$, the maximum tree depth is constrained to $d = 8$, and the number of boosting iterations

is fixed at $M = 600$. An L2 regularisation strength of $\lambda = 1.0$ is applied to mitigate overfitting. The HGB classifier is trained exclusively on the frozen Boruta-selected features within each outer cross-validation fold.

2) Support Vector Machine (SVM):

A linear Support Vector Machine (SVM) is trained on the LASSO-selected feature subset $\mathcal{S}_{\text{LASSO}}$. Prior to classification, features are standardised to zero mean and unit variance. The classifier solves the soft-margin primal optimisation problem:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|_2^2 + C_{\text{SVM}} \sum_{i=1}^n \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)), \quad (6)$$

where $\mathbf{w}$ and b define the separating hyperplane, and C_{SVM} controls the trade-off between margin maximisation and classification error. A linear kernel is used with $C_{\text{SVM}} = 1.0$, and class imbalance is addressed via balanced class weights.

The decision function $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$ is passed directly to the meta-learner, avoiding the need for probability calibration within the stacking framework.

3) k -Nearest Neighbors (KNN):

A distance-weighted K -Nearest Neighbours (KNN) classifier is included as a complementary non-parametric base learner. Unlike the HGB and SVM branches, this pathway operates on the full descriptor space without explicit feature selection, capturing local neighbourhood structure across all available molecular descriptors.

Prior to classification, features are standardised to zero mean and unit variance to ensure meaningful distance computations. Given a query sample $\mathbf{x}$, KNN identifies the set $\mathcal{N}_k(\mathbf{x})$ of its k nearest neighbours under Euclidean distance. The class probability is estimated as:

$$P(y = 1 \mid \mathbf{x}) = \frac{\sum_{i \in \mathcal{N}_k(\mathbf{x})} w_i \mathbb{I}(y_i = 1)}{\sum_{i \in \mathcal{N}_k(\mathbf{x})} w_i}, \quad (7)$$

where $w_i = 1/d(\mathbf{x}, \mathbf{x}_i)$ and $d(\cdot, \cdot)$ denotes Euclidean distance. The number of neighbours is set to $k = 5$, with distance weighting to emphasise closer neighbours.

This branch complements the global decision boundaries learned by HGB and SVM by exploiting local similarity structure.

D. Stacking Ensemble (Meta-Learner)

The predictions of the three base learners are combined using a stacking ensemble. Let $\hat{s}_m(\mathbf{x})$ denote the output score of the m -th base learner, where scores correspond to class probabilities (HGB, KNN) or decision-function values (SVM). Out-of-fold predictions are generated via stratified 5-fold cross-validation and stacked into a meta-feature vector:

$$\mathbf{z}(\mathbf{x}) = [\hat{s}_{\text{HGB}}(\mathbf{x}), \hat{s}_{\text{SVM}}(\mathbf{x}), \hat{s}_{\text{KNN}}(\mathbf{x})]. \quad (8)$$

A logistic regression meta-learner with L2 regularisation is trained on these meta-features by minimising the regularised log-loss:

$$\min_{\boldsymbol{\beta}} \left[- \sum_{i=1}^n \log \sigma(\beta_0 + \mathbf{z}_i^\top \boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}\|_2^2 \right], \quad (9)$$

where $\sigma(\cdot)$ denotes the sigmoid function. The regularisation strength is set to $C_{\text{meta}} = 1.0$ (corresponding to $\lambda = 1/C_{\text{meta}} = 1.0$), and class imbalance is addressed via balanced class weights.

The meta-learner adaptively reweights the contributions of individual base learners, amplifying informative branches while suppressing weaker ones. To prevent information leakage, meta-training is performed exclusively on out-of-fold predictions generated from strictly partitioned cross-validation folds.

E. Evaluation Framework

Model performance is evaluated using a stratified nested cross-validation protocol to ensure unbiased generalisation estimates. An outer stratified 5-fold cross-validation is used for performance evaluation, while all feature selection, base learner training, and stacking operations are confined to the corresponding training folds. Boruta feature selection is performed once per outer training fold and subsequently frozen to prevent information leakage.

In addition, a stratified 80/20 train-test split is used for final model training and figure generation. Performance is assessed using ROC-AUC, F1-score, and precision-recall metrics, which are appropriate for imbalanced QSAR classification. All models are implemented using scikit-learn. All performance metrics are computed on held-out data only.

IV. MODEL ANALYSIS

1) *Overall Performance:* The proposed ensemble demonstrates strong predictive performance on the independent test set. Using a stratified 80/20 split comprising 211 samples with a positive class rate of 0.336, the model achieves a ROC-AUC of 0.941 and an F1-score of 0.844.

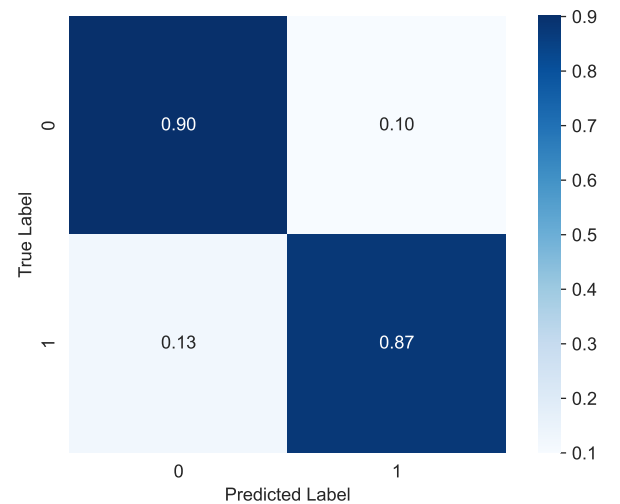


Fig. 2: Normalised confusion matrix on the independent test set. Class 0: Not Ready Biodegradable (NRB); Class 1: Ready Biodegradable (RB).

Figure 2 presents the normalised confusion matrix. The model exhibits balanced classification behaviour, achieving

a recall of 0.873 for the positive class while maintaining a precision of 0.816. Performance on the negative class remains high, with a precision of 0.933 and recall of 0.900, indicating effective control of false positives.

$$\text{Sensitivity} = \frac{TP}{TP + FN} = 0.87 \quad (10)$$

$$\text{Specificity} = \frac{TN}{TN + FP} = 0.90 \quad (11)$$

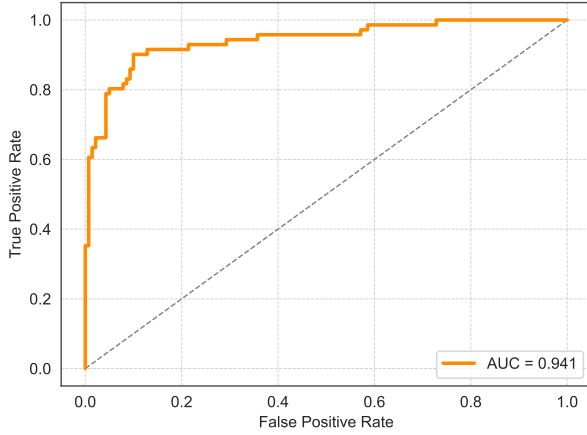


Fig. 3: Receiver operating characteristic (ROC) curve of the stacked ensemble.

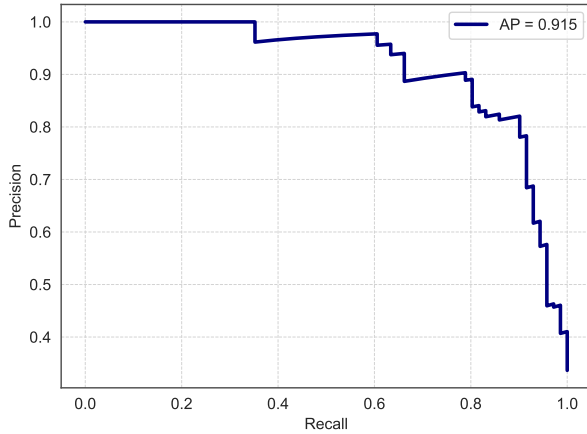


Fig. 4: Precision–recall (PR) curve of the stacked ensemble.

Discriminative ability is illustrated by the ROC curve in Figure 3, where the high AUC confirms strong class separability. Figure 4 further highlights precision–recall performance, which is particularly informative under class imbalance and confirms robust minority class detection.

A. Base Learner Comparison

To quantify the contribution of each branch, the base learners were evaluated independently on the same test set. Table I summarises the results.

While the HGB classifier achieves the highest AUC (0.947), it exhibits lower sensitivity (0.831), indicating a tendency

TABLE I: Performance comparison of base learners and stacked ensemble on the independent test set.

Model	AUC	F1	Sens.	Spec.
Stacked Ensemble	0.941	0.844	0.873	0.900
HGB (Boruta)	0.947	0.831	0.831	0.914
SVM (LASSO)	0.915	0.775	0.873	0.807
KNN (Full)	0.917	0.816	0.845	0.886

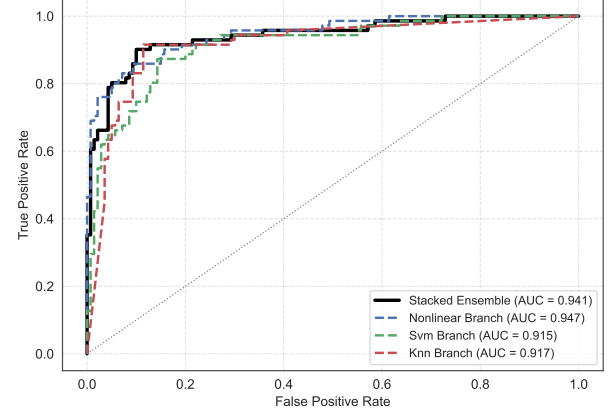


Fig. 5: ROC curves comparing individual base learners and the stacked ensemble.

to miss biodegradable compounds. The stacked ensemble achieves the best F1 score and the most balanced trade-off between sensitivity and specificity. The SVM branch, operating on only 15 LASSO-selected features, achieves competitive sensitivity but lower specificity, while the KNN branch provides stable performance using the full feature space.

Figure 5 visualises the ROC curves of each branch alongside the stacked ensemble, confirming that the heterogeneous learners capture complementary decision boundaries.

B. Learning Curve Analysis

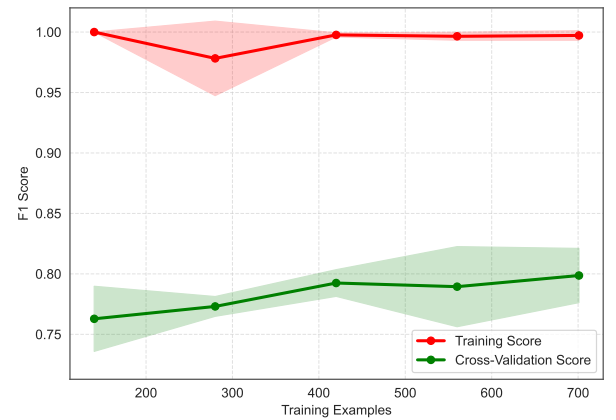


Fig. 6: Learning curve showing training and cross-validation F1 scores as a function of training set size.

Figure 6 shows the learning curve of the architecture used. Training performance remains consistently high across all training sizes, with mean F1 scores ranging from 0.98 to

1.00. This near-perfect training performance is characteristic of expressive ensemble methods [16] and reflects the combined capacity of gradient boosting, support vector machines, and instance-based learning. Importantly, this high training score does not indicate overfitting, as evidenced by the stable validation performance.

Validation performance improves steadily as the training set size increases, rising from an F1-score of approximately 0.76 at 168 samples to approximately 0.80 at 702 samples. This positive slope indicates that the model continues to benefit from additional training data.

The generalisation gap decreases from approximately 0.24 to 0.20 as training size increases. According to classical learning theory [17], a narrowing gap with increasing data is indicative of a model with appropriate complexity for the available sample size. The relatively small gap at the largest training size suggests that the L2-regularised meta-learner and pathway-specific feature selection effectively control model variance.

The shaded regions in Figure 6 represent ± 1 standard deviation across cross-validation folds. The narrow bands for training scores indicate stable fitting behaviour, while the slightly wider bands for validation scores reflect the inherent variability of held-out evaluation on limited data. This variability is consistent with observations on similarly sized QSAR datasets [18].

Notably, the validation curve does not plateau, suggesting that the model has not yet reached its asymptotic performance. This observation aligns with recommendations in the QSAR literature that biodegradation models benefit from expanded and diverse training sets [19].

C. Meta-Learner Interpretation

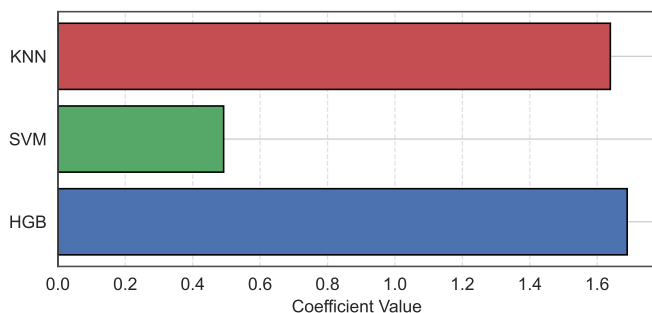


Fig. 7: Coefficients of the logistic regression meta-learner for each base learner.

In stacked generalisation, the meta-learner combines base learner outputs to produce a final prediction [9]. The L2-regularised logistic regression employed provides a transparent linear combination while mitigating collinearity between correlated predictions.

Figure 7 visualises the meta-learner coefficients. All three base learners receive positive, non-zero weights, confirming that each branch contributes to the final prediction. The HGB branch receives the largest coefficient (1.65), reflecting its strong individual performance and the discriminative power of

Boruta-selected features. The KNN branch receives a similarly high coefficient (1.62), suggesting that local neighbourhood information provides complementary signal not captured by global decision boundaries. The SVM branch receives a lower coefficient (0.42); however, in ensemble learning, weaker learners can still improve overall performance if their errors are uncorrelated with those of stronger learners [8].

The moderate ratio between coefficients (approximately 4:1) indicates differentiation without extreme dominance. L2 regularisation enforces this balance by penalising large coefficient magnitudes and distributing weight across learners. The relative ordering ($HGB \approx KNN > SVM$) remained consistent across cross-validation folds, indicating stable meta-learner behaviour.

V. CONCLUSION AND RECOMMENDATIONS

This study presents a heterogeneous stacking framework for QSAR biodegradation classification that integrates complementary feature-selection strategies with diverse learning paradigms. The architecture combines non-linear feature relevance detection via Boruta, sparse linear selection via LASSO, and full-space distance-based learning, unified through an L2-regularised meta-learner. Crucially, all feature selection and preprocessing steps are embedded within a nested cross-validation framework to prevent information leakage—a common but often overlooked source of optimistic bias in QSAR modelling [18].

The proposed ensemble achieves a ROC-AUC of 0.941 and an F1-score of 0.844 on an independent test set, with balanced sensitivity (0.873) and specificity (0.900). These results compare favourably with previously reported models on the UCI biodegradation dataset [19]. Learning curve analysis confirms low variance and effective regularisation, while base learner comparisons demonstrate that the stacked ensemble achieves superior balance across metrics compared to any individual branch.

Several limitations warrant consideration. First, the framework was validated on a single dataset; generalisation to structurally diverse chemical spaces remains to be established. Second, the current architecture relies on pre-computed molecular descriptors, which may not capture all relevant structural features. Third, the computational cost of nested cross-validation with Boruta feature selection may limit scalability to larger datasets.

Future QSAR work could extend this framework in several directions. Alternative molecular representations, such as graph neural network embeddings or extended connectivity fingerprints, may encode structural information not captured by traditional descriptors [20]. The meta-learning component could also be replaced with non-linear alternatives, such as gradient boosting or shallow neural networks, to model more complex interactions between base learner outputs. Additionally, evaluation on external datasets with different chemical spaces would help establish the generalisability of the approach to broader QSAR applications.

REFERENCES

- [1] E. Benfenati and G. Gini, “Computer models for reach: the case of biodegradability,” *SAR and QSAR in Environmental Research*, vol. 20, no. 3-4, pp. 197–202, 2009.

- [2] A. Cherkasov, E. N. Muratov, D. Fourches, A. Varnek, I. I. Baskin, M. Cronin, J. Dearden, P. Gramatica, Y. C. Martin, R. Todeschini *et al.*, "Qsar modeling: where have you been? where are you going to?" *Journal of Medicinal Chemistry*, vol. 57, no. 12, pp. 4977–5010, 2014.
- [3] R. Todeschini and V. Consonni, *Molecular Descriptors for Chemoinformatics*. Wiley, 2009.
- [4] D. Stumpfe and J. Bajorath, "Exploring the structure–activity landscape," *Wiley Interdisciplinary Reviews: Computational Molecular Science*, vol. 2, no. 4, pp. 538–548, 2012.
- [5] UCI Machine Learning Repository, "Qsar biodegradation dataset," 2013. [Online]. Available: <https://archive.ics.uci.edu/ml/datasets/QSAR+biodegradation>
- [6] W. M. S. Russell and R. L. Burch, *The Principles of Humane Experimental Technique*. Methuen, 1959.
- [7] A. Tropsha, "Recent advances in qsar-based profiling," *Drug discovery today: technologies*, vol. 1, no. 4, pp. 349–356, 2004.
- [8] T. G. Dietterich, "Ensemble methods in machine learning," *Multiple Classifier Systems*, vol. 1857, pp. 1–15, 2000.
- [9] D. Wolpert, "Stacked generalization," *Neural Networks*, 1992.
- [10] L. Breiman, "Stacked regressions," *Machine Learning*, vol. 24, no. 1, pp. 49–64, 1996.
- [11] M. B. Kursa and W. R. Rudnicki, "Feature selection with the boruta package," *Journal of Statistical Software*, vol. 36, no. 11, pp. 1–13, 2010.
- [12] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [13] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [14] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21–27, 1967.
- [15] R. Tibshirani, "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 58, no. 1, pp. 267–288, 1996.
- [16] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2009.
- [17] V. N. Vapnik, *Statistical Learning Theory*. Wiley, 1998.
- [18] A. Tropsha, "Best practices for QSAR model development, validation, and exploitation," *Molecular Informatics*, vol. 29, no. 6-7, pp. 476–488, 2010.
- [19] K. Mansouri, T. Ringsted, D. Ballabio, R. Todeschini, and V. Consonni, "Quantitative structure–activity relationship models for ready biodegradability of chemicals," *Journal of Chemical Information and Modeling*, vol. 53, no. 4, pp. 867–878, 2013.
- [20] Z. Wu, B. Ramsundar, E. N. Feinberg, J. Gomes, C. Geniesse, A. S. Pappu, K. Leswing, and V. Pande, "Moleculenet: A benchmark for molecular machine learning," *Chemical Science*, vol. 9, no. 2, pp. 513–530, 2018.